# 203733: porphyritic granitic rock, Great Eastern mine

## *(Lawlers Tonalite, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)*

## Location and sampling

Leonora (SH 51-1), Wildara (3041)
MGA Zone 51, 258596E 6891742N

Sampled on 2 September 2020

This sample was collected from the 125.0 – 126.3 m depth interval of drillcore EMSD1399, drilled in 2015–16 by Agnew Gold Mining Company Pty Ltd. The drillhole is located at the Great Eastern mine, about 6.9 km east-northeast of Mount McAuley, 5.7 km south of Agnew Bluff, and 2.2 km east of the Old Agnew Road.

## Geological context

The unit sampled is the Lawlers Tonalite of the Kalgoorlie Terrane (Sapkota et al., 2020). The Lawlers Tonalite consists of multiple intrusions, mainly metamorphosed tonalite, quartz diorite and granodiorite, that intruded mafic–ultramafic extrusive and intrusive units in the core of the Lawlers Anticline (Foden et al., 1984; Beardsmore, 2002; Duuring et al., 2012; Hayman et al., 2015). This sample was collected to refine the age of the Lawlers Tonalite. A tonalite about 10.4 km to the southwest provided an igneous crystallization age of 2691 ± 9 Ma and an age of 2809 ± 11 Ma for an inherited zircon component (sample SONG05, Thébaud et al., 2018). A monzogranite gneiss, about 4.9 km to the southwest, yielded an igneous crystallization age of 2690 ± 6 Ma (sample PHAG029, Thébaud et al., 2018). An igneous crystallization age of 2628 ± 21 Ma was reported by Thébaud et al. (2018) for a monzogranite (sample PHAG030) about 4.2 km to the northwest, although the published data suggest that a more appropriate age of 2677 ± 19 Ma (MSWD = 2.2) can be based on the five oldest analyses with discordance <5%. Metagranodiorite about 9.6 km to the southwest yielded dates of 2808 ± 7 and 2696 ± 7 Ma (GSWA 182454, Wingate et al., 2018a), and a metatonalite dyke intruded into greenstones about 12.0 km to the north yielded an igneous crystallization age of 2674 ± 3 Ma (GSWA 182455, Wingate et al., 2018b).

## Petrographic description

The sample is a medium- to coarse-grained, porphyritic granitic rock (Fig. 1). K-feldspar (orthoclase and microcline) is subhedral to euhedral and zoned. Rare plagioclase phenocrysts are up to 8 mm in size and surrounded by a fine-grained mosaic of quartz and K-feldspar. Feldspars are slightly altered to epidote and sericite. Fine-grained biotite forms aggregates and discontinuous stringers that wrap around some of the larger feldspar grains. Accessory minerals include titanite, pyrite and zircon. Biotite is partially altered to chlorite, and granular aggregates of epidote are associated with the ferromagnesian minerals.

## Zircon morphology

Zircons isolated from this sample are colourless to dark brown and mainly subhedral. The crystals are up to 160 µm long, and equant to elongate, with aspect ratios up to 4:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and many crystals contain high-U, metamict zones. A CL image of all zircons is shown in Figure 2.

## Analytical details

This sample was analysed on 9–10 and 15–16 October 2020, using SHRIMP-B. Analyses 1.1 to 8.1 (spot numbers 1–8) were obtained during the first session, together with 13 analyses of the M257 standard, of which 12 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.24% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the first session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.39%. Analyses 9.1 to 3.3 (spot numbers 9–18) were obtained during the second session, together with 16 analyses of the M257 standard, of which 14 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.28% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

## Results

Eighteen analyses were obtained from 11 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

## Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Twelve analyses are >5% discordant. The dates obtained from these 12 analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining six analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

**Figure 1. Drillcore image for sample 203733: porphyritic granitic rock, Great Eastern mine**

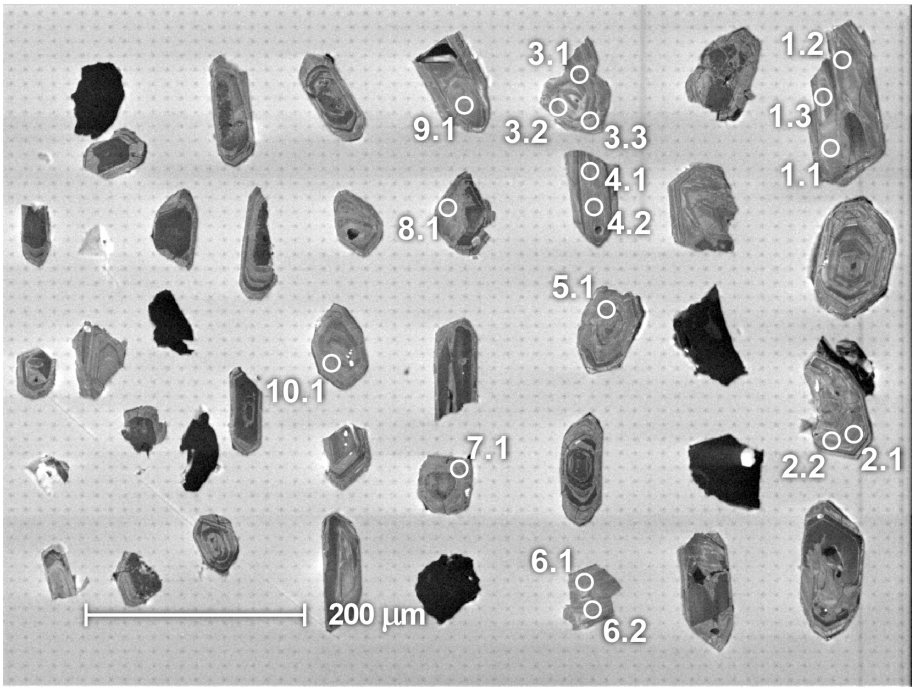


**Figure 2. Cathodoluminescence image of all zircons from sample 203733: porphyritic granitic rock, Great Eastern mine. Numbered circles indicate the approximate locations of analysis sites**

**Table 1. Ion microprobe analytical results for zircons from sample 203733: porphyritic granitic rock, Great Eastern mine**

| Group ID | Spot no. | Grain. spot | $^{238}U$ (ppm) | $^{232}Th$ (ppm) | $^{232}Th/^{238}U$ | f204 (%) | $^{238}U/^{206}Pb$ ± 1σ | | $^{207}Pb/^{206}Pb$ ± 1σ | | $^{238}U/^{206}Pb^*$ ± 1σ | | $^{207}Pb^*/^{206}Pb^*$ ± 1σ | | $^{238}U/^{206}Pb^*$ date (Ma) ±1σ | | $^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ | | Disc. (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P | 15 | 6.2 | 439 | 554 | 1.26 | 0.027 | 2.091 | 0.038 | 0.17688 | 0.00185 | 2.091 | 0.038 | 0.17664 | 0.00186 | 2519 | 38 | 2622 | 17 | 3.9 |
| P | 4 | 4.1 | 282 | 260 | 0.92 | 0.099 | 2.115 | 0.023 | 0.17761 | 0.00133 | 2.117 | 0.024 | 0.17673 | 0.00138 | 2494 | 23 | 2622 | 13 | 4.9 |
| I | 3 | 3.1 | 360 | 401 | 1.12 | -0.012 | 2.038 | 0.021 | 0.18174 | 0.00122 | 2.038 | 0.021 | 0.18185 | 0.00122 | 2574 | 23 | 2670 | 11 | 3.6 |
| I | 13 | 1.2 | 288 | 170 | 0.59 | 0.020 | 2.025 | 0.025 | 0.18214 | 0.00121 | 2.025 | 0.025 | 0.18196 | 0.00122 | 2587 | 27 | 2671 | 11 | 3.1 |
| I | 2 | 2.1 | 283 | 180 | 0.64 | 0.128 | 2.005 | 0.023 | 0.18320 | 0.00129 | 2.008 | 0.023 | 0.18206 | 0.00134 | 2605 | 24 | 2672 | 12 | 2.5 |
| I | 1 | 1.1 | 204 | 118 | 0.58 | 0.132 | 2.044 | 0.025 | 0.18437 | 0.00147 | 2.047 | 0.025 | 0.18319 | 0.00154 | 2565 | 26 | 2682 | 14 | 4.4 |
| D | 9 | 9.1 | 616 | 1239 | 2.01 | 0.193 | 3.156 | 0.031 | 0.13451 | 0.00076 | 3.162 | 0.031 | 0.13281 | 0.00088 | 1772 | 16 | 2135 | 12 | 17.0 |
| D | 12 | 4.2 | 773 | 1442 | 1.87 | 0.146 | 2.802 | 0.059 | 0.13646 | 0.00294 | 2.806 | 0.059 | 0.13516 | 0.00296 | 1965 | 36 | 2166 | 38 | 9.3 |
| D | 11 | 11.1 | 627 | 510 | 0.81 | 0.172 | 2.885 | 0.234 | 0.14569 | 0.00858 | 2.890 | 0.234 | 0.14417 | 0.00860 | 1915 | 134 | 2278 | 103 | 15.9 |
| D | 8 | 8.1 | 664 | 384 | 0.58 | 0.082 | 2.525 | 0.033 | 0.15384 | 0.00158 | 2.527 | 0.033 | 0.15311 | 0.00159 | 2150 | 24 | 2381 | 18 | 9.7 |
| D | 10 | 10.1 | 525 | 257 | 0.49 | 0.084 | 2.397 | 0.024 | 0.15672 | 0.00079 | 2.399 | 0.024 | 0.15597 | 0.00083 | 2246 | 19 | 2412 | 9 | 6.9 |
| D | 7 | 7.1 | 644 | 816 | 1.27 | 0.014 | 2.343 | 0.021 | 0.15864 | 0.00100 | 2.343 | 0.021 | 0.15851 | 0.00101 | 2292 | 18 | 2440 | 11 | 6.1 |
| D | 6 | 6.1 | 370 | 627 | 1.69 | -0.033 | 2.295 | 0.023 | 0.16176 | 0.00112 | 2.294 | 0.023 | 0.16205 | 0.00113 | 2332 | 20 | 2477 | 12 | 5.9 |
| D | 17 | 1.3 | 435 | 438 | 1.01 | -0.058 | 2.232 | 0.025 | 0.17260 | 0.00176 | 2.230 | 0.025 | 0.17312 | 0.00177 | 2388 | 22 | 2588 | 17 | 7.7 |
| D | 14 | 2.2 | 352 | 304 | 0.87 | 0.081 | 2.395 | 0.028 | 0.17655 | 0.00120 | 2.397 | 0.028 | 0.17583 | 0.00126 | 2248 | 22 | 2614 | 12 | 14.0 |
| D | 5 | 5.1 | 342 | 244 | 0.71 | 0.068 | 2.170 | 0.023 | 0.17730 | 0.00124 | 2.172 | 0.023 | 0.17669 | 0.00127 | 2442 | 22 | 2622 | 12 | 6.9 |
| D | 16 | 3.2 | 237 | 253 | 1.07 | 0.095 | 2.109 | 0.042 | 0.17987 | 0.00130 | 2.111 | 0.042 | 0.17903 | 0.00137 | 2500 | 42 | 2644 | 13 | 5.4 |
| D | 18 | 3.3 | 176 | 89 | 0.51 | 0.115 | 2.165 | 0.032 | 0.18064 | 0.00166 | 2.168 | 0.032 | 0.17961 | 0.00176 | 2445 | 30 | 2649 | 16 | 7.7 |

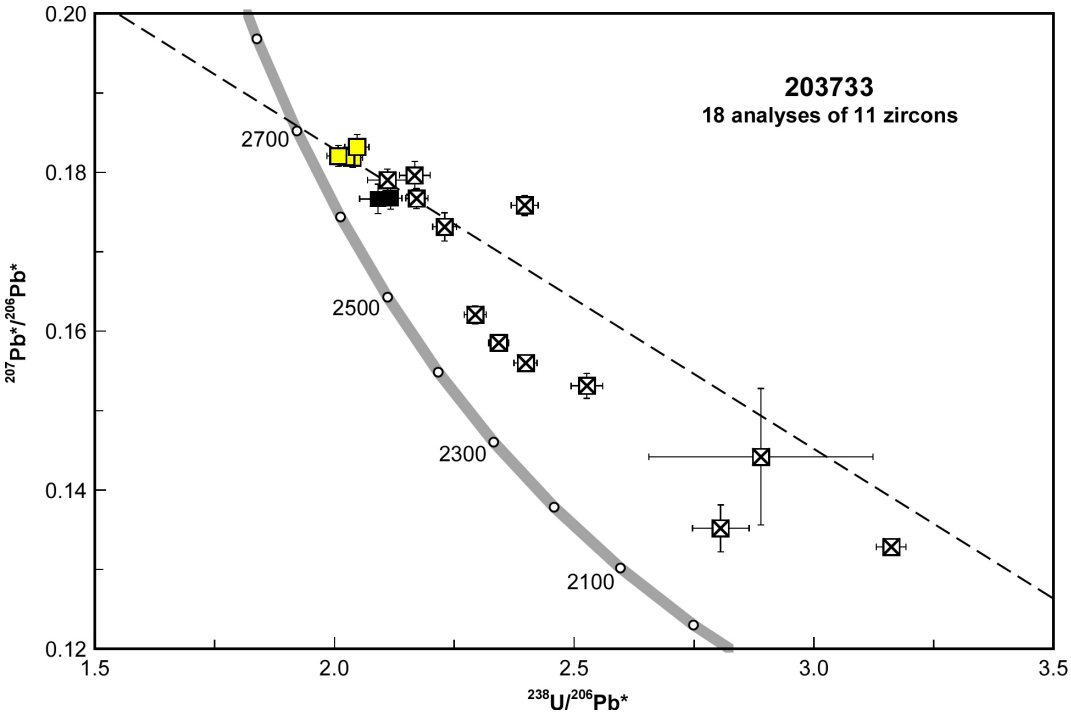


**Figure 3. U–Pb analytical data for sample 203733: porphyritic granitic rock, Great Eastern mine. Yellow squares indicate Group I (magmatic zircons); black squares indicate Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >5%). The dashed line indicates a discordia regression through data in Groups I and P and selected data in Group D**

Group I comprises four analyses of three zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2673 ± 12 Ma (MSWD = 0.18).

Group P comprises two analyses of two zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of c. 2622 Ma.

The date of 2673 ± 12 Ma for the four analyses in Group I is 3.4% discordant, and therefore interpreted as a minimum age for igneous crystallization. The dates of c. 2622 Ma for the two analyses in Group P are interpreted to reflect loss of radiogenic Pb.

A discordia regression (Fig. 3) through 10 analyses in Groups I, P and D that indicate dates >2500 Ma and are <8% discordant yields concordia intercepts at 2718 ± 42 and c. 1400 Ma (MSWD = 0.77). The upper intercept of 2718 ± 42 Ma may overestimate the crystallization age, because the distribution of all data indicates the zircons have been affected by more than one episode of Pb loss.

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